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(54) METHODS AND DEVICES FOR ASEPTIC DRY TRANSFER

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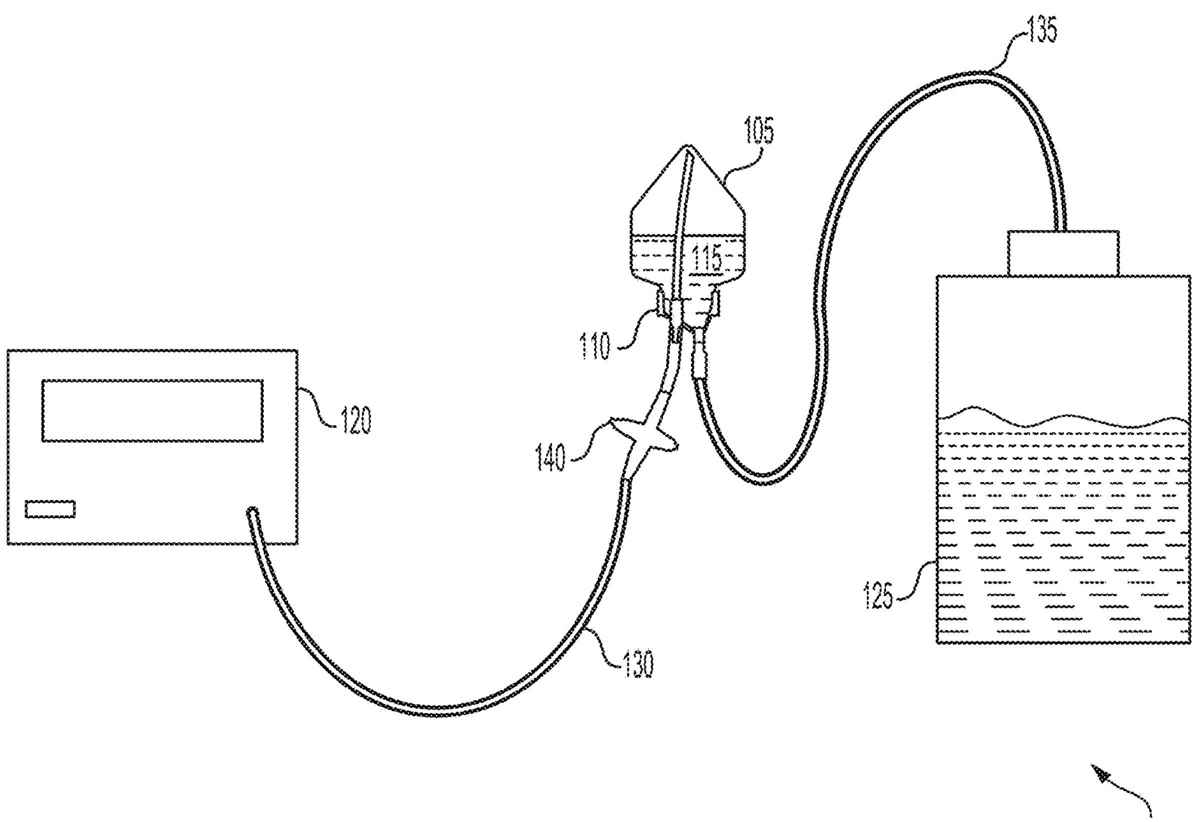
(57) **ABSTRACT**

Devices and systems for product transfer, such as a dissolvable microcarriers, are described. In one example, the product delivery device may include an inlet port; a conical section that may include a wide end and a narrow end; an outlet port flush with the narrow end of the conical section and extending away from the conical section; and a securement feature configured to connect the wide end of the conical section to a container. In another example, a system for aseptic dry product delivery may include a container at least partially filled with the aseptic dry product; a product delivery device; a pressure source connected to an inlet port; and a receiving vessel where the aseptic dry product is collected.



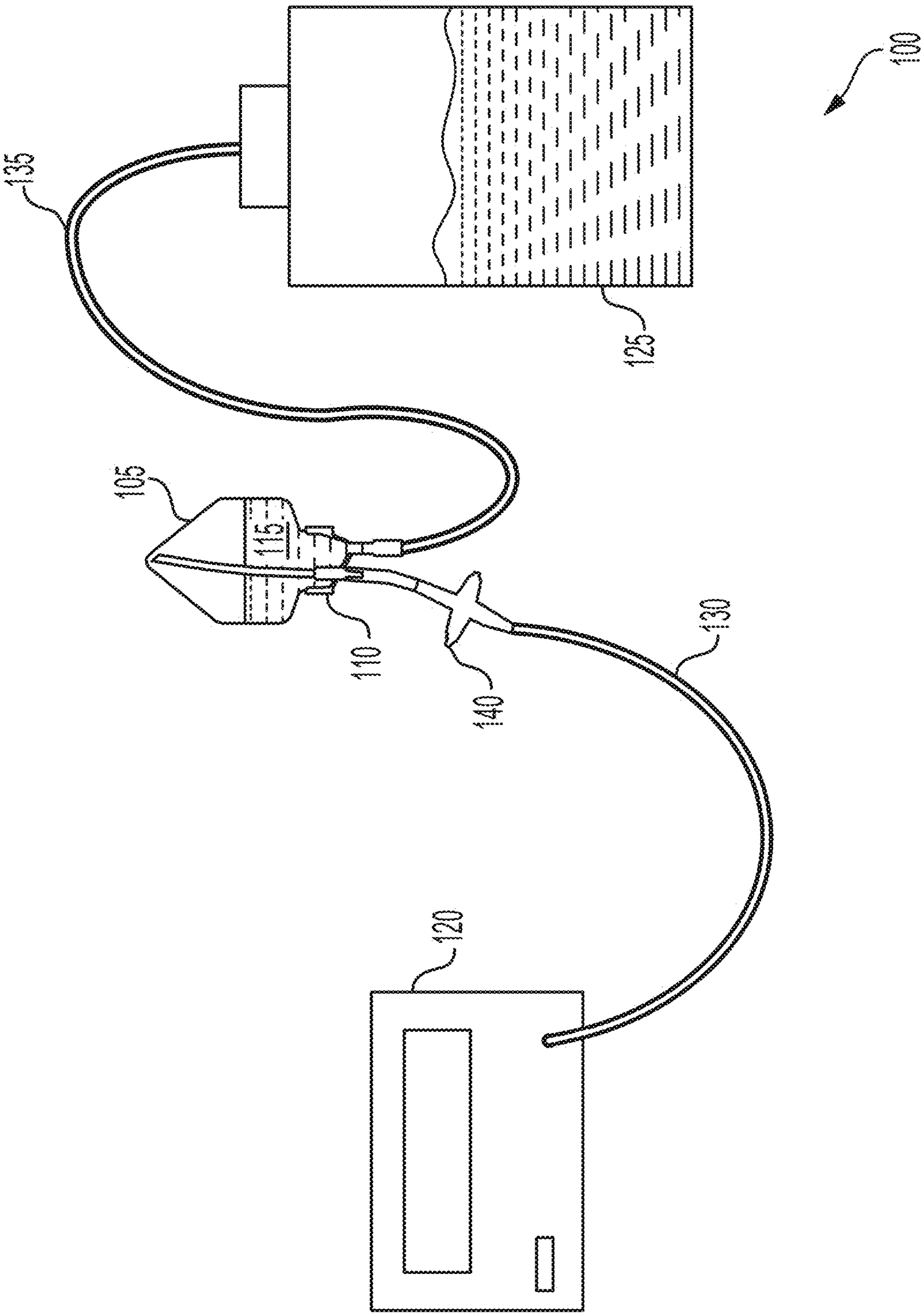


FIG. 1

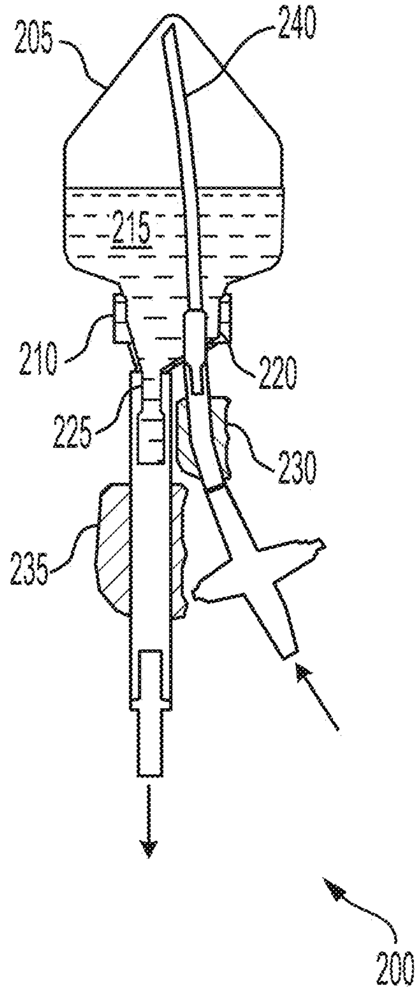


FIG. 2

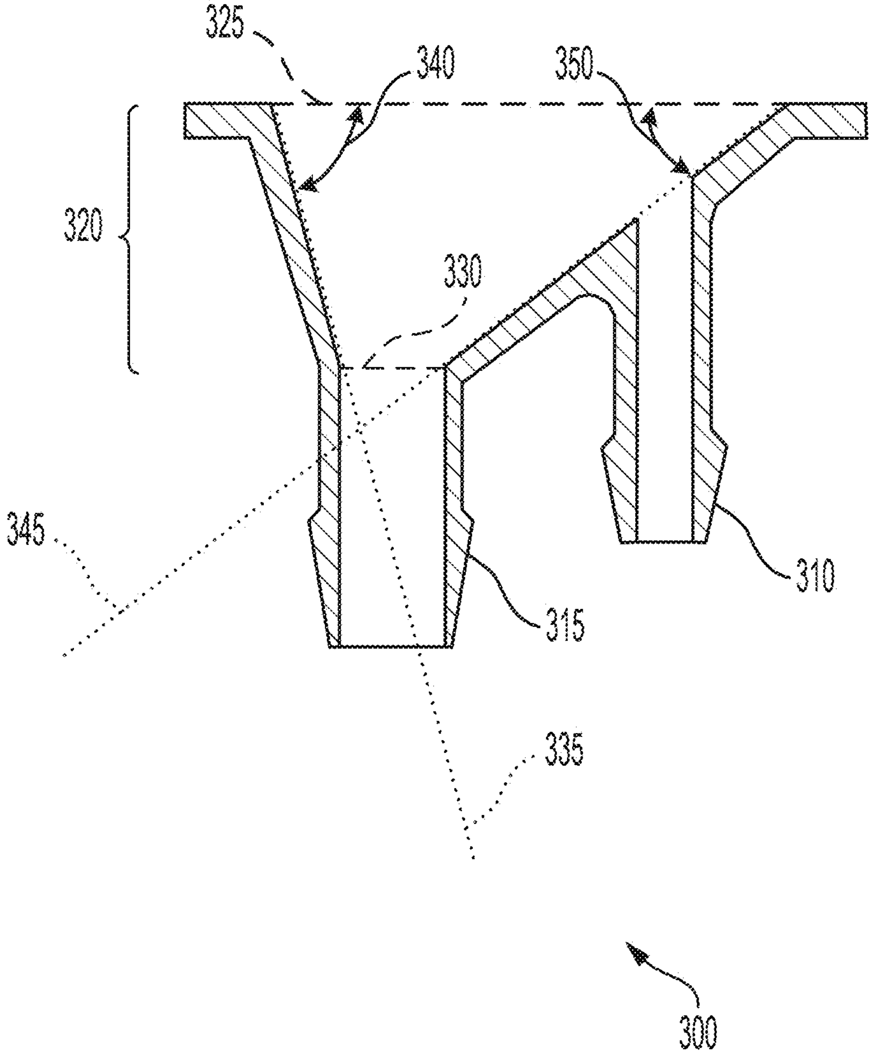


FIG. 3

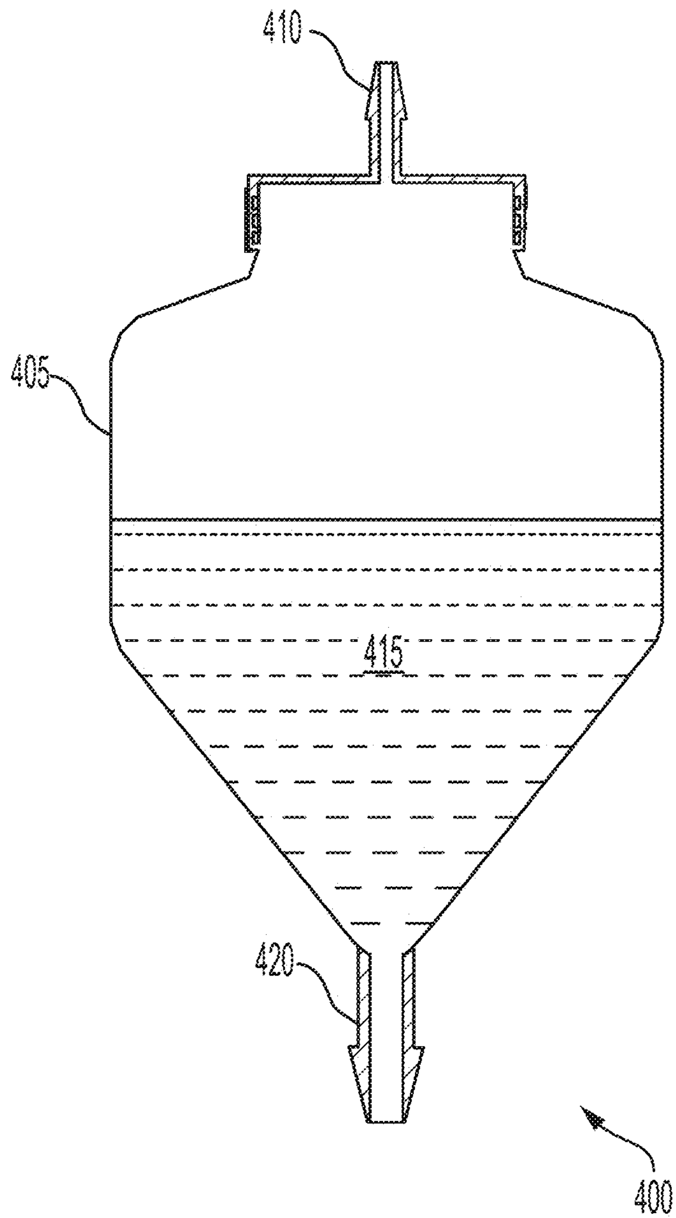


FIG. 4

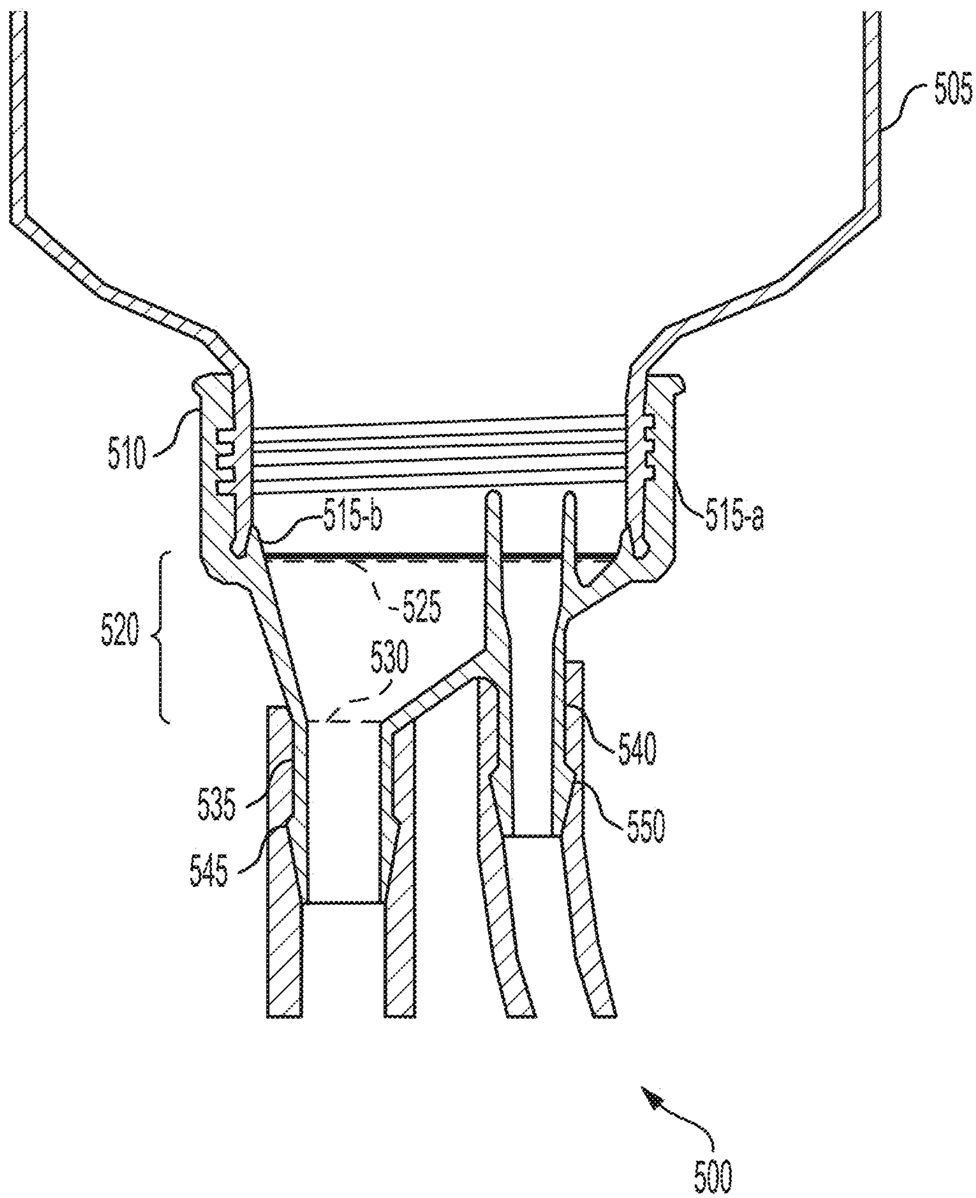


FIG.5

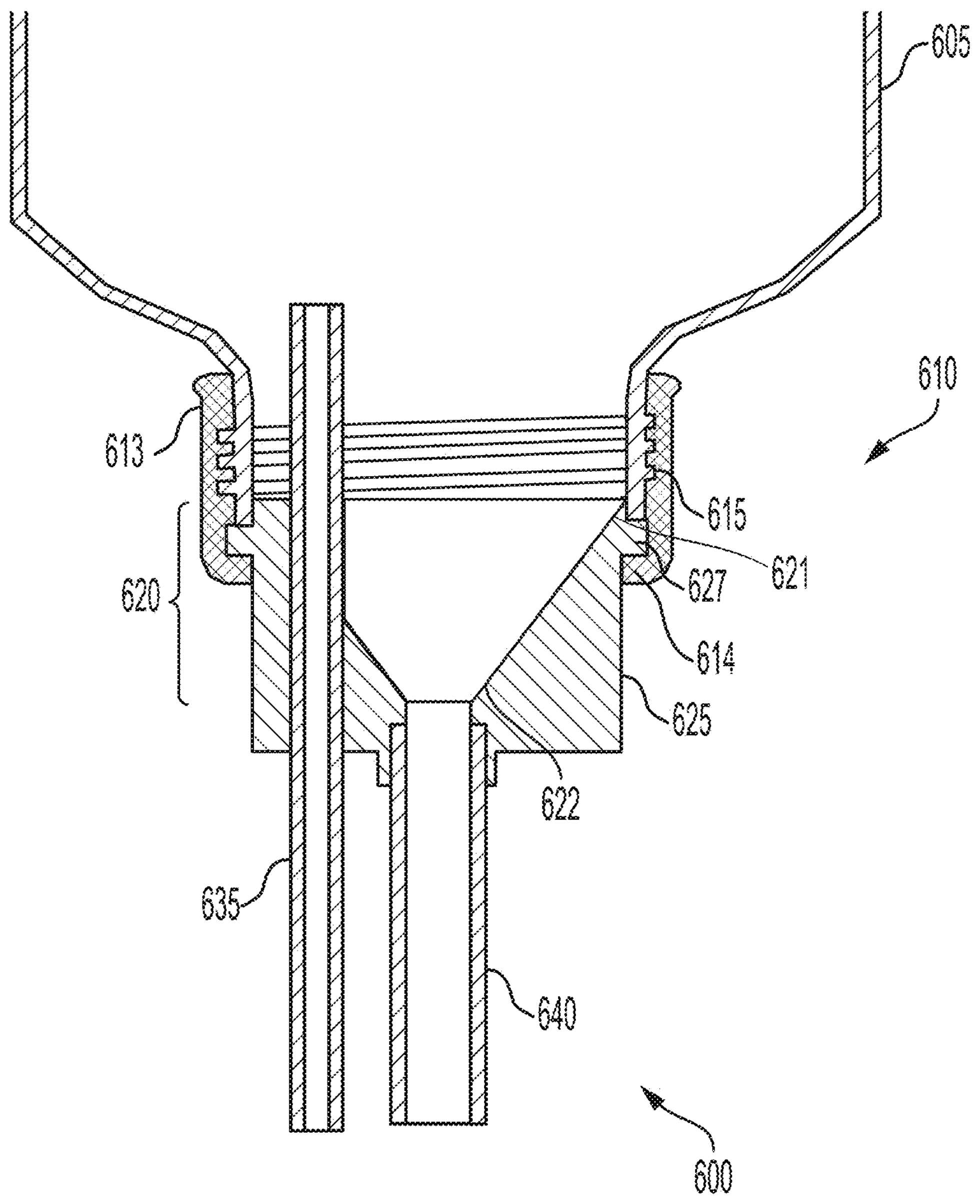


FIG.6

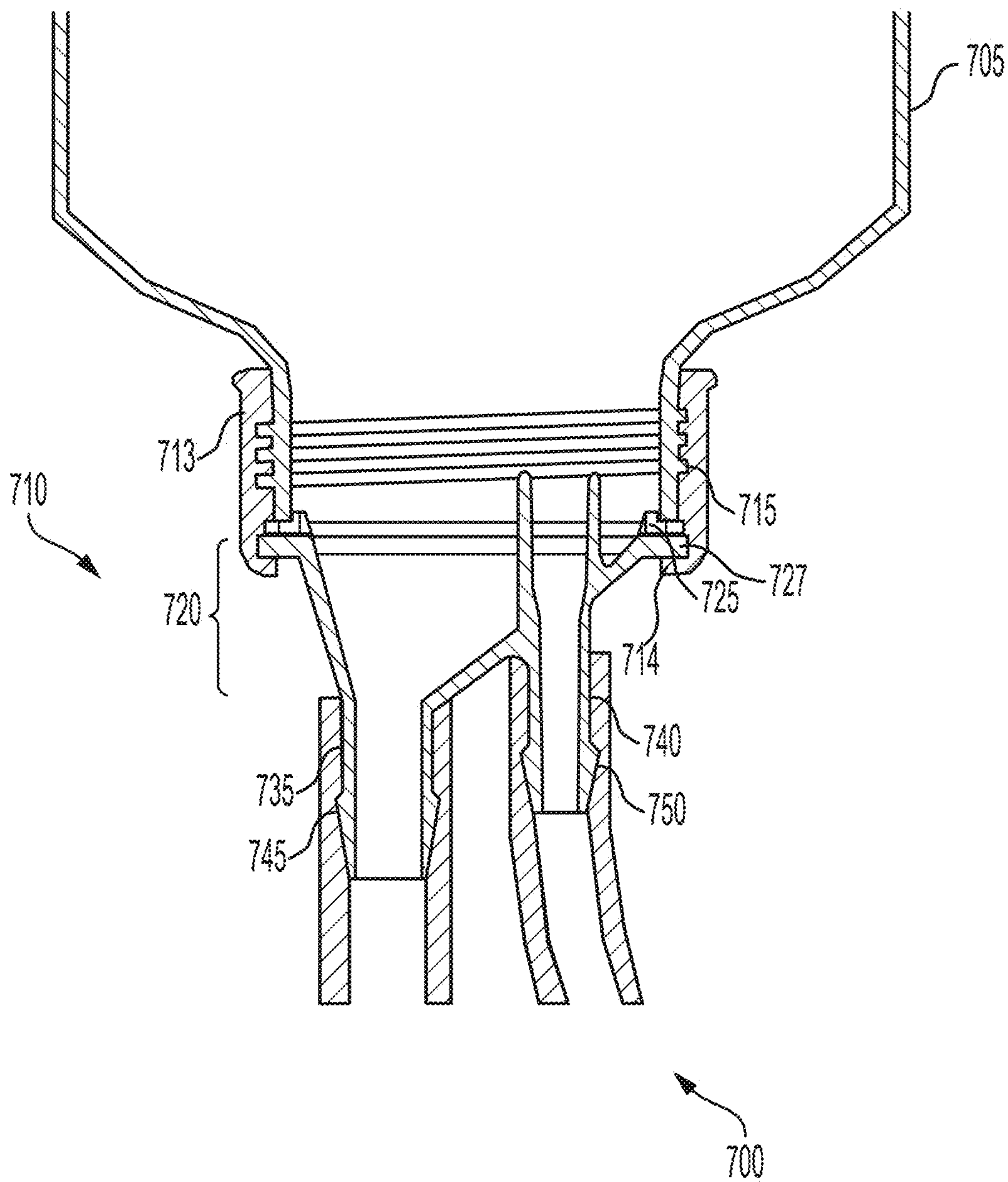


FIG.7

METHODS AND DEVICES FOR ASEPTIC DRY TRANSFER

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is a divisional of U.S. patent application Ser. No. 18/143,239 filed on May 4, 2023 which is a divisional of U.S. patent application Ser. No. 16/766,028 filed on May 21, 2020 which was granted as U.S. Pat. No. 11,655,440 issued on May 23, 2023 and which claims the benefit of priority to the national stage application under 35 U.S.C. § 371 of International Application No. PCT/US2018/062470 filed on Nov. 26, 2018, which claims the benefit of priority under 35 U.S.C § 120 of U.S. Provisional Application Ser. No. 62/592,709 filed on Nov. 30, 2017, the contents of which are relied upon and incorporated herein by reference in their entirety.

BACKGROUND

[0002] The following relates generally to product transfer, and more specifically to transferring dry product in an aseptic system.

[0003] Products, such as dry powders, may be difficult to transfer out of a container based on the properties of the container. For example, dry powders may not completely transfer out of a system due to product hang up. Powders may get hung up on rough or flat surfaces, which leads to a loss of usable product. Dry powders, such as dissolvable microcarriers, may be transported in a sealed container to prevent contamination from the environment. In many cases, cell growth or regulatory standards may require a closed, aseptic, or sterile environment.

[0004] The deployment of dissolvable microcarriers in adherent cell cultures allows more surface area per unit volume for cell growth than two dimensional surfaces. Traditionally, cell growth within bioreactors was limited by the surface area available on traditional two-dimensional adherent surfaces. Microcarriers may have the ability to easily scale up cell growth within bioreactors. Dry microcarriers may have a substantially spherical shape with a diameter of about 30 microns. Once hydrated, microcarriers may have a diameter of about 250 microns.

[0005] Complete product transfer may be significant for the cost and yields desired by a consumer. Product that remains in the container may be unusable and can result in the loss of money and a lower process yield. To enable increased product transfer in systems such as aseptic cell cultures with microcarriers, an aseptic transfer device may be needed to introduce product (e.g., the microcarriers) into the system (e.g., culture vessel).

SUMMARY

[0006] The present disclosure is directed towards a device and systems for product delivery. In some embodiments, the product delivery device can include an inlet port, a conical section including a wide end and a narrow end, an outlet port flush with the narrow end of the conical section and extending away from the conical section, and a securement feature configured to connect the wide end of the conical section to a container.

[0007] In some embodiments, the diameter of the narrow end is smaller than the diameter of the wide end. In some embodiments, walls of the conical section extend from the wide end to the narrow end and can define an angle of the conical section. In some cases, the angle of the conical section can vary along a circumference of the narrow end, the wide end, or both. In other cases, the angle of the conical section can be uniform along a circumference of the narrow end, the wide end, or both.

[0008] In some embodiments, the device or system can further include a single piece cap that can include the inlet port, the conical section, the outlet port, and the securement feature. According to various aspects of the disclosure, the securement feature can be configured to form a wedge fit with the container. In some cases, the cap can be formed by injection molding.

[0009] In some embodiments, the device or system can further include a two-piece cap that can include the inlet port, the conical section, the outlet port, and the securement feature. According to various aspects of the disclosure, the device or system can also include a first piece comprising a stopper, and a second piece comprising the securement feature, wherein the first piece can engage with the second piece to create a seal when connected to the container.

[0010] In some embodiments, the device or system can further include a three or more-piece cap including the inlet port, the conical section, the outlet port, and the securement feature. According to various aspects of the disclosure, the device or system can also include a first piece including the conical section and the outlet port, a second piece including the securement feature, and a third piece including a gasket, wherein the first piece engages with the second piece to deform the gasket when connected to the container.

[0011] In some embodiments, the gasket can include an internal angle based at least in part on an angle of the conical section. In some embodiments, the securement feature includes threads configured to engage an internal surface of the container.

[0012] In some embodiments, the product delivery device can include at least one of polyethylene, polypropylene, polystyrene, polycarbonate, or silicon. In some embodiments, the inlet port and outlet port include an external barb.

[0013] A system for product delivery can include a container at least partially filled with the product, and a product delivery device including a conical section with a wide end and a narrow end, and an outlet port flush with the narrow end of the conical section, wherein the product is transferred from the conical section through the outlet port.

[0014] In some embodiments, the device or system can further include an inlet port, and a securement feature configured to connect the wide end of the conical section to the container. According to various aspects of the disclosure, the device or system can also include a dip tube connected to the inlet port and extending to an opposite side of the container that the inlet port.

[0015] In some embodiments, the device or system can further include a gasket between the product delivery device and the container, the gasket configured to deform to create a hermetic seal. In some embodiments, the device or system can be aseptic.

[0016] In some embodiments, the product is a dry free flowing powder. In some cases, the product is dissolvable microcarriers.

[0017] Another system for product delivery can include a container at least partially filled with the aseptic dry product, a product delivery device including a conical section with a wide end and a narrow end, and an outlet port flush with the

narrow end of the conical section. According to his system, the dry product is transferred from the conical section through the outlet port, a pressure source connected to an inlet port, and a receiving vessel, where the aseptic dry product is collected.

[0018] In some embodiments, the device or system can further include a gasket between the product delivery device and the container, the gasket being configured to deform to create a hermetic seal. In some embodiments, the device or system can be closed to the environment. In some embodiments, the device or system can be aseptic or sterile.

[0019] In some embodiments, the device or system can further include a dip tube connected to the inlet port that extends into the container. The dip tube can be configured to transfer pressure from the pressure source into the container.

[0020] In some embodiments, the product is a dry, free-flowing powder. In some cases, the product includes dissolvable microcarriers. In some examples, the dry product can be hydrated in the receiving vessel. In some embodiments, the inlet port and outlet port include an external barb.

[0021] In some embodiments, the device or system can further include a first tube connected to the inlet port, a first clamp configured to pinch the first tube, a second tube connected to the outlet port, and a second clamp configured to pinch the second tube.

[0022] One or more representative embodiments is provided to illustrate the various features, characteristics, and advantages of the disclosed subject matter. The embodiments are provided in the context of glass electrochemical sensors. It should be understood, however, that many of the concepts can be used in a variety of other settings, situations, and configurations. For example, the features, characteristics, advantages, etc., of one embodiment can be used alone or in various combinations and sub-combinations with one another.

[0023] The Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. The Summary and the Background are not intended to identify key concepts or essential aspects of the disclosed subject matter, nor should they be used to constrict or limit the scope of the claims. For example, the scope of the claims should not be limited based on whether the recited subject matter includes any or all aspects noted in the Summary and/or addresses any of the issues noted in the Background.

BRIEF DESCRIPTION OF DRAWINGS

[0024] The preferred and other embodiments are disclosed in association with the accompanying drawings in which:

[0025] FIG. **1** illustrates a cross-sectional view of an example of a closed system for product transfer in accordance with various aspects of the present disclosure.

[0026] FIG. **2** illustrates a cross-sectional view of an example of a product delivery system in accordance with various aspects of the present disclosure.

[0027] FIG. **3** illustrates a cross-sectional view of an example of a product delivery device in accordance with various aspects of the present disclosure.

[0028] FIG. **4** illustrates a cross-sectional view of an example of an alternative product delivery system in accordance with various aspects of the present disclosure.

[0029] FIG. **5** illustrates a cross-sectional view of an example of a single piece product delivery system in accordance with various aspects of the present disclosure.

[0030] FIG. **6** illustrates a cross-sectional view of an example of a two-piece product delivery system in accordance with various aspects of the present disclosure.

[0031] FIG. **7** illustrates a cross-sectional view of an example of a three-piece product delivery system in accordance with various aspects of the present disclosure.

DETAILED DESCRIPTION

[0032] The present disclosure relates to devices and systems for delivering materials (e.g., dry products) wherein the devices and systems enable as complete of a transfer as possible, including, but in no way limited to, the aseptic transfer of dissolvable microcarriers (DMCs). An easy to use container can enable the optimal delivery of product while minimizing the product lost, which can be hung up in the container (e.g., the lid and outlet port of the container). Such a device can be valuable for systems that require specific environmental standards, such as a clean room, to avoid contamination. The transfer can enable the system to remain closed to the outside environment, allowing reduced requirements for the external environment of the system. In some cases, product that is sterile or aseptic could remain sterile or aseptic using the devices and systems described herein. The devices and systems described can also be designed to withstand abnormal pressure (e.g., above or below atmospheric pressure). A pressure gradient in combination with the described devices and systems can allow a more complete delivery of product. In some cases, the product to be delivered by this device can be dry or wet, and the product can be a fluid or a solid.

[0033] Conventional systems for delivering materials (e.g., dry products) can be unable to minimize the residual material in a container due to the geometry of such devices. The present disclosure relates to a product delivery system that enables improved product transfer by reducing resistance of the path for a product to exit the container in the system, which allows for minimized product hang up. This disclosure lends itself to the transfer of DMCs in a closed system, thus enabling the process of adherent cell growth in a bioreactor. In particular, this system can include a container at least partially filled with product (e.g., aseptic dry DMCs), and a cap device that connects to the container. In some examples, this process also allows for the use of back pressure to force more product out of the container and allows for a more rapid flow out of the container, relative to the use of gravity alone, and through any associated tubing. The product can leave the container through an outlet port that is flush with a narrow end of a conical section. The outlet port may not extend into the conical section, which reduces the resistance of the path for a product to exit the container. Although this disclosure can be described in relation to the transfer of DMCs, the devices and systems can be applied to the transfer of a variety of materials.

[0034] FIG. **1** illustrates a cross-sectional view of an example of a closed system **100** for product transfer in accordance with various aspects of the present disclosure. As illustrated in FIG. **1**, the system **100** can include a container **105**, a product delivery device **110**, a pressure source **120**, and a receiving vessel **125**.

[0035] The container **105** can be at least partially filled with a product **115** (e.g., a free-flowing powder such as dry DMCs). The container **105**, as shown in FIG. **1**, is inverted. Before the container **105** is inverted, a dip tube can be inserted in the inlet port, the product delivery device **110**

(e.g., a cap) can be secured onto the container **105**, and the tubing **130** and **135** can be connected to the inlet port and outlet port, respectively. The container **105** can be secured in an inverted position by a clamp or other commonly known configurations for positioning a device.

[0036] The product delivery device **110** can be secured onto the container **105** (e.g., screwing onto internal or external threads on the container), formed as an integral part of the container **105** by any number of manufacturing methods, including, but in no way limited to, 3D printing, blow molding, and the like. The product delivery device **110** can have a conical internal shape, where internal is defined herein as being disposed between the walls of the device **110**. The conical section of the product delivery device **110** can allow for optimized transfer of product **115** to the receiving vessel **125** by reducing residual product in the container **105**.

[0037] The pressure source **120** can be connected to the inlet of the product delivery device **110** by tubing **130**, which can include a vent **140** and vent filter to catch or otherwise prevent contaminates from entering the container **105**. Alternatively, the vent **140** can be attached to the container **105**, rather than the tubing **130**. In any event, the tubing **130** and vent **140** form a fluid flow path between the pressure source **120** and the inlet of the container **105**. The pressure source **120** can provide pressure through the tubing **130** to the dip tube in the container **105** in order to provide pressure at the top of the container **105**. As illustrated in FIG. **1**, the top of the container **105** is shown with no product **115**. Consequently, the addition of pressure to the container **105** can apply a force on the product **115** to move to a lower pressure area (e.g., the receiving vessel **125**). Thus, the product **115** will travel out of the container **105** and have minimal, if any, product hang up in the container **105** due to the shape of the conical section and the placement of the outlet port.

[0038] Receiving vessel **125** can be connected to the outlet of the product delivery device **110** by tubing **135**. In some examples, the receiving vessel **125** can be a bioreactor where the product **115** transferred from the container **105** can aid in cell growth. If the product **115** were DMCs, then the DMCs can be hydrated within the receiving vessel **125**.

[0039] The entire system **100** can be closed and be configured to maintain an aseptic or sterile environment. The size of the container **105**, product delivery device **110**, and other aspects of the system **100** can be a variety of sizes, based on the amount of the product **115**.

[0040] FIG. **2** illustrates a cross-sectional view of an example of a product delivery system **200** in accordance with various aspects of the present disclosure. As illustrated in FIG. **2**, the system **200** can include a container **205**, a product delivery device **210**, product **215**, an inlet port **220**, and an outlet port **225**. The container **205** can be an example of container **105** as shown in FIG. **1**. Product delivery device **210** can be an example of product delivery device **110**.

[0041] The inlet port **220** of the product delivery device **210** can be connected to a dip tube **240** that extends from the inlet port **220** to the opposite side of the container **205** (e.g., the top of the container **205**). A pressure source (**120** from FIG. **1**) can be connected to the inlet port **220** outside of the container **205** and can provide pressure through the dip tube **240** into the container **205** in order to provide pressure at the top of the container **205** where the top of the container **205** is shown with no product **215**. The addition of pressure to the container **205** can apply a force on the product **215** to move to a lower pressure area. Thus, the product **215** will travel out of the container **205** and have minimal if any product hang up in the container **205** due to the shape of the conical section and the placement of the outlet port **225**.

[0042] In some cases, the tubing that is fluidly connected to the inlet port **220** can include a clamp **230**. The clamp **230** can temporarily engage the tubing (e.g., tubing **130** in FIG. **1**) to create a pinch in the tubing to prevent product **215** from flowing further into the tubing. Another clamp **235** can be placed on tubing that is fluidly connected to the outlet port **225**. Clamp **235** can be similar to clamp **230**, and clamp **235** can temporarily engage the tubing (e.g., tubing **135** in FIG. **1**) to create a pinch in the tubing to prevent product **215** from flowing further into the tubing. Once the system is ready for operation, the clamps **230** and **235** can be disengaged to allow product, pressurized air, or other elements to flow through the tubing. For example, the clamps **230** and **235** can be disengaged after tubing on the far side of the clamp, the side opposite the outlet port **225**, is welded to other tubing in the system and evacuated of any undesirable elements. These clamps **230** and **235** can allow for cleaner tube welds and for maintaining the seal of the system **200** from the external environment. Thus, the system **200** can remain closed.

[0043] According to various aspects of this disclosure, the outlet port **225** can be arranged on the product delivery device **210** such that the uppermost portion of the outlet port **225** is flush with the interior conical section of the container **205**. For example, the edge between the conical section and the outlet port **225** can be continuous and smooth such that there is reduced resistance for a product **215** to exit the container **205** through the outlet port **225**.

[0044] FIG. **3** illustrates a cross-sectional view of an example of a product delivery device **300**, in accordance with various aspects of the present disclosure. As illustrated in FIG. **3**, the product delivery device **300** can include an inlet port **310**, an outlet port **315**, and a conical section **320**.

[0045] Inlet port **310** can be arranged anywhere on the device **300** apart from and physically separated from the outlet port **315**. In some examples, the inlet port **310** can be connected to the conical section **320** at a distance from the outlet port **315**. As illustrated in FIG. **3**, the inlet port **310** can be flush with the conical section **320**. In other cases, the inlet port **310** can extend into the conical section **320** and can include a dip tube disposed therein.

[0046] The outlet port **315** can be attached to the narrow end **330** of the conical section. The outlet port **315** can be flush with the conical section **320** (e.g., a continuous surface with a corner). The outlet port **315** can be positioned anywhere on the device **300** provided that the outlet port **315** is flush with the narrow end of the conical section.

[0047] Conical section **320** can also include a wide end **325** and a narrow end **330**. The narrow end **330** can have a diameter that is smaller than the diameter of the wide end **325**. Dotted lines **335** and **345** are shown to clarify the angles of the conical section **320** walls. Angles **340** and **350** are depicted between the plane of the wide end **325** and dotted lines **345** and **355**, respectively. Angles **340** and **350** can be different or the same. If the angles **340** and **350** are different, the angles around the entire conical section can vary along the circumference of the wide end **325**, the narrow end **330**, or both. If the angles **340** and **350** are the same, the angles around the entire conical section can be uniform along the circumference of the wide end **325** and the narrow end **330**.

Angles **340** and **350** can be greater than 0 degrees, for example greater than 25 degrees.

[0048] FIG. **4** illustrates a cross-sectional view of an example of an alternative product delivery system **400** in accordance with various aspects of the present disclosure. As illustrated in FIG. **4**, system **400** can include a container **405**, inlet port **410**, and outlet port **420**. This system **400** can be implemented in system **100**.

[0049] Container **405** can be at least partially filled with product **415**. In some cases, container **405** can have a conical shape as illustrated in FIG. **4**.

[0050] Inlet port **410** can be secured to the top of the container **405**. In some examples, inlet port **410** can allow pressure to be transferred into the container **405** to apply a pressure to the product **415**. The pressure can force the product **415** out of the container **405** through the outlet port **420**.

[0051] Outlet port **420** can be connected to the conical section of the container **405**. The outlet port **420** can be flush with the narrow end of the conical section of the container **405**.

[0052] FIG. **5** illustrates a cross-sectional view of an example of a single piece product delivery system **500**, in accordance with various aspects of the present disclosure. As illustrated in FIG. **5**, system **500** can include a container **505** and a product delivery device **510**.

[0053] Product delivery device **510** can be a single piece of material that includes securement features **515**, a conical section **520**, an outlet port **535**, and an inlet port **540**. Product delivery device **510** can be formed by any number of manufacturing methodologies including, but in no way limited to, injection molding. In some examples, the product delivery device **510** can be manufactured of any suitable material, including, but in no way limited to, at least one of polyethylene, polypropylene, polystyrene, polycarbonate, or silicone.

[0054] Securement features **515**-*a* and **515**-*b* can attach the wide end **525** of the conical section **520** to the container **505**. In some cases, securement feature **515**-*a* can engage with threads of the container **505**. As shown, the securement features **515**-*a* of the container **505** are illustrated as external threads formed as circumferential protrusions on the lower outer surface of the container **505** adjacent to the bottommost opening in the container, where they can be rotationally engaged with internal threads formed on the inner circumferential surface of the product delivery device. However, in some cases, the securement feature **515**-*a* can engage with internal threads of the container **505**. Securement feature **515**-*b* can form a wedge or interference fit with the inside or top of the container mouth. The securement features **515**-*b* can have an internal angle similar in geometry to the conical section **520** to allow smooth flow of product. The securement features **515**-*a* and **515**-*b* allow for an improved seal between the product delivery device **510** and the container **505**. In some cases, the seal can allow the system **500** to be closed.

[0055] Conical section **520** can include a wide end **525** and a narrow end **530**. An outlet port **535** can be flush with the narrow end **530**. As illustrated, outlet port **535** and inlet port **540** can include one or more external barbs **545** and **550**, respectively. Barbs **545** and **550** allow for tubing to securely connect to each port. In some cases, the barbs **545** and **550** allow tubing to remain connected to the ports when the system is pressurized.

[0056] FIG. **6** illustrates a cross-sectional view of an example of a two-piece product delivery system **600** in accordance with various aspects of the present disclosure. As illustrated in FIG. **6**, system **600** can include a container **605** and a two-piece product delivery device **610**.

[0057] Product delivery device **610** can be two separate component pieces, combined to form the delivery device **610**. The first piece can include a collar piece **613** including a distal lip **614**, having the securement features **615** formed on an inner surface of the collar piece **613**, as illustrated in FIG. **6**. The second piece of the product delivery device **610** can include the conical section **620**, a stopper **625** including a seating lip **627** that engages the distal lip **614** of the collar piece **613**, an inlet port **635**, and an outlet port **640**. In some examples, the product delivery device **610** can be formed of at least one or more of polyethylene, polypropylene, polystyrene, polycarbonate, or silicone.

[0058] Securement features **615** can attach the wide end of the conical section **620** to the container **605**. In some cases, securement feature **615** can engage with threads of the container **605**. As shown, the threads are formed as protrusions circumferentially formed on the external wall of the bottommost opening of the container **605**, however, in some cases, the securement feature **615** can engage with internal threads of the container **605**. When the securement features **615** of the first piece, such as the collar **613**, engage with the threads of the container **605**, the second piece can be positioned between the first piece and the container's mouth such that the edges of the stopper **625** are secured. More specifically, according to one embodiment, the seating lip **627** of the stopper **625** engages the distal lip **614** of the collar piece **613**, and is compressed between the distal lip **614** of the collar piece and the end of the container **605** as it is advanced along the securement features. This configuration allows for a compressed seal between the product delivery device **610** and the container **605**. In some cases, the seal can allow the system **600** to be closed.

[0059] Conical section **620** can include a wide end **621** and a narrow end **622**, which can be hollowed out of the stopper **625**. Stopper **625** can be made of a flexible material (e.g., silicone), and the collar piece **613** can be made of a rigid material (e.g., polyethylene). The angle of the conical section **620** can be the same or differ along the circumference of the wide end **621**, the narrow end **622**, or both. If the angles are the same around the entire conical section, they can be uniform along the circumference of the wide end and the narrow end, as shown. In some cases, the angles can be greater than 0 degrees, for example greater than 25 degrees.

[0060] An outlet port **640** can be flush with the narrow end of the conical section **620**. As illustrated, inlet port **635** and outlet port **640** can extend away from the container **605**. The outlet port **640** can be centered within the stopper **625**. Although not shown, the inlet port **635** and outlet port **640** can include external barbs.

[0061] FIG. **7** illustrates a cross-sectional view of an example of a three-piece product delivery system **700** in accordance with various aspects of the present disclosure. As illustrated in FIG. **7**, the system **700** can include a container **705** and a three-piece product delivery device **710**.

[0062] The three-piece product delivery device **710** can be formed of three separate and distinct pieces, which can be manufactured of the same or different types of material. The first piece of the three-piece product delivery device **710** can include a collar piece **713** including a distal lip **714**, and a securement feature **715** formed on an inner surface thereof, as illustrated in FIG. **7**. The second piece of the three-piece product delivery device **710** can include conical section **720**, an outlet port **735**, and an inlet port **740**. As illustrated in FIG. **7**, the second piece includes a seating lip **727** formed on an upper periphery of the conical section **720**, the seating lip **727** being configured to interface with the distal lip **714** of the collar piece **713** when assembled. The third piece of the three-piece product delivery device **710** can include a gasket **725**. As illustrated in FIG. **7**, the gasket **725** engages the seating lip **727** of the second piece and is compressed between the top end of the container **705**, the seating lip **727** of the conical section **720**, and the distal lip **714** of the collar piece **713**. In some examples, the product delivery device **710** can be manufactured of any combination of materials including, but in no way limited to, polyethylene, polypropylene, polystyrene, polycarbonate, and/or silicone.

[0063] As illustrated in FIG. **7**, securement features **715** can operate to secure the wide end of the conical section **720** to the container **705**. In some cases, securement feature **715** can engage with threads of the container **705**. As shown, the threads are formed as protrusions circumferentially formed on the external wall of the bottommost opening of the container, however, in some cases, the securement feature **715** can engage with internal threads of the container **705**. When the securement features **715** of the collar **713** engage with the threads of the container **705**, the seating lip **727** of the second piece and the gasket **725** of the third piece can be positioned between the distal lip **714** of the collar **713** of the first piece and the mouth of the container **705**, such that the intermediate elements are compressed, thereby sealing the interface between the various pieces. The gasket **725** can be made of a flexible material that can be deformed when force is added. The gasket **725** in the system **700** can be deformed to create a seal between the container **705** and the conical section **720**. The gasket **725** can have an internal angle similar in geometry to the conical section **720** to allow smooth flow of product. The securement features **715** and gasket **725** allow for an improved seal between the product delivery device **710** and the container **705**. In some cases, the seal can allow the system **700** to be closed.

[0064] The conical section **720** can include a wide end and a narrow end. An outlet port **735** can be flush with the narrow end. As illustrated, the outlet port **735** and the inlet port **740** can include one or more external barbs **745** and **750**, respectively. Barbs **745** and **750** allow for tubing to securely connect to each port. In some cases, the barbs **745** and **750** can be important when the system is pressurized to allow tubing to remain connected to the ports.

[0065] According to an aspect (1) of the present disclosure, a product delivery device is provided. The product delivery device comprises: an inlet port; a conical section including a wide end and a narrow end; an outlet port disposed flush with the narrow end of the conical section extending away from the conical section; and a securement feature formed on the product delivery device, wherein the securement feature is configured to couple the wide end of the conical section to a container.

[0066] According to an aspect (2) of the present disclosure, the product delivery device of aspect (1) is provided, wherein the diameter of the narrow end is smaller than the diameter of the wide end.

[0067] According to an aspect (3) of the present disclosure, the product delivery device of aspect (2) is provided, wherein walls of the conical section extend from the wide end to the narrow end and define an angle of the conical section.

[0068] According to an aspect (4) of the present disclosure, the product delivery device of aspect (4) is provided, wherein, wherein the angle of the conical section varies along a circumference of the narrow end or the wide end.

[0069] According to an aspect (5) of the present disclosure, the product delivery device of aspect (3) is provided, wherein the angle of the conical section is uniform along a circumference of the narrow end or the wide end.

[0070] According to an aspect (6) of the present disclosure, the product delivery device of any of aspects (1)-(5) is provided, further comprising a single piece cap including the inlet port, the conical section, the outlet port, and the securement feature.

[0071] According to an aspect (7) of the present disclosure, the product delivery device of aspect (6) is provided, wherein, wherein the securement feature is configured to form a compression fit with the container.

[0072] According to an aspect (8) of the present disclosure, the product delivery device of aspect (6) is provided, wherein the cap is formed by injection molding.

[0073] According to an aspect (9) of the present disclosure, the product delivery device of any of aspects (1)-(8) is provided, further comprising a two-piece cap including the inlet port, the conical section, the outlet port, and the securement feature.

[0074] According to an aspect (10) of the present disclosure, the product delivery device of aspect (9) is provided, further comprising: a first piece including a stopper; and a second piece including the securement feature, wherein the first piece engages with the second piece to create a seal when connected to the container.

[0075] According to an aspect (11) of the present disclosure, the product delivery device of any of aspects (1)-(10) is provided, further comprising a three-piece cap including the inlet port, the conical section, the outlet port, and the securement feature.

[0076] According to an aspect (12) of the present disclosure, the product delivery device of aspect (11) is provided, further comprising: a first piece comprising the inlet port, the conical section and the outlet port; a second piece comprising the securement feature; and a third piece comprising a gasket, wherein the first piece engages with the second piece to deform the gasket when connected to the container.

[0077] According to an aspect (13) of the present disclosure, the product delivery device of aspect (12) is provided, wherein the gasket comprises an internal angle based at least in part on an angle of the conical section.

[0078] According to an aspect (14) of the present disclosure, the product delivery device of any of aspects (1)-(13) is provided, wherein the securement feature comprises threads configured to engage an external surface of the container.

[0079] According to an aspect (15) of the present disclosure, the product delivery device of any of aspects (1)-(14) is provided, wherein the product delivery device comprises at least one of polyethylene, polypropylene, polystyrene, polycarbonate, or silicone.

[0080] According to an aspect (16) of the present disclosure, the product delivery device of any of aspects (1)-(15) is provided, wherein the inlet port and outlet port comprise an external barb.

[0081] According to an aspect (17) of the present disclosure, a system for product delivery is provided. The system for product delivery comprises: a container; and a product delivery device comprising: a conical section with a wide end and a narrow end; and an outlet port flush with the narrow end of the conical section, wherein the product is transferred from the conical section through the outlet port.

[0082] According to an aspect (18) of the present disclosure, the system for product delivery of aspect (17) is provided, further comprising: an inlet port; and a securement feature configured to connect the wide end of the conical section to the container.

[0083] According to an aspect (19) of the present disclosure, the system for product delivery of aspect (18) is provided, further comprising a dip tube connected to the inlet port and extending into the container to a side opposite the inlet port.

[0084] According to an aspect (20) of the present disclosure, the system for product delivery of aspect (18) is provided, further comprising a gasket between the product delivery device and the container, the gasket configured to deform to create a hermetic seal.

[0085] According to an aspect (21) of the present disclosure, the system for product delivery of any of aspects (17)-(20) is provided, wherein the container further comprises a product, the product including a dry free flowing powder.

[0086] According to an aspect (22) of the present disclosure, the system for product delivery of aspect (21) is provided, wherein the product comprises dissolvable microcarriers.

[0087] According to an aspect (23) of the present disclosure a system for aseptic dry product delivery is provided. The system for aseptic dry product delivery comprises: a container; a product delivery device including: a conical section with a wide end and a narrow end; and an outlet port disposed flush with the narrow end of the conical section, wherein the dry product is transferred from the conical section through the outlet port; a pressure source connected to an inlet port; and a receiving vessel configured to receive the aseptic dry product.

[0088] According to an aspect (24) of the present disclosure, the system for product delivery of aspect (23) is provided, further comprising a gasket disposed between the product delivery device and the container, the gasket configured to deform to create a hermetic seal within the system.

[0089] According to an aspect (25) of the present disclosure, the system for product delivery of any of aspects (23)-(24) is provided, wherein the system is closed to the environment.

[0090] According to an aspect (26) of the present disclosure, the system for product delivery of any of aspects (23)-(25) is provided, wherein a dip tube connected to the inlet port extends into the container, and wherein the dip tube is configured to transfer pressure from the pressure source into the container.

[0091] According to an aspect (27) of the present disclosure, the system for product delivery of any of aspects (23)-(26) is provided, wherein the inlet port and outlet port comprise an external barb.

[0092] According to an aspect (28) of the present disclosure, the system for product delivery of any of aspects (23)-(27) is provided, further comprising: a first tube connected to the inlet port; a first clamp configured to pinch the first tube; a second tube connected to the outlet port; and a second clamp configured to pinch the second tube.

[0093] According to an aspect (29) of the present disclosure, the system for product delivery of any of aspects (23)-(28) is provided, wherein the container is at least partially filled with an aseptic dry product, the dry product comprising dissolvable microcarriers.

[0094] According to an aspect (30) of the present disclosure, the system for product delivery of aspect (29) is provided, wherein the dry product is hydrated in the receiving vessel.

[0095] It should be appreciated that some components, features, and/or configurations can be described in only one embodiment, but these same components, features, and/or configurations can be applied or used in or with many other embodiments and should be considered applicable to the other embodiments, unless stated otherwise or unless such a component, feature, and/or configuration is technically impossible to use with the other embodiment. Thus, the components, features, and/or configurations of the various embodiments can be combined in any manner and such combinations are expressly contemplated and disclosed by this statement.

[0096] The terms recited in the claims should be given their ordinary and customary meaning as determined by reference to relevant entries in widely used general dictionaries and/or relevant technical dictionaries, commonly understood meanings by those in the art, etc., with the understanding that the broadest meaning imparted by any one or combination of these sources should be given to the claim terms (e.g., two or more relevant dictionary entries should be combined to provide the broadest meaning of the combination of entries, etc.) subject only to the following exceptions: (a) if a term is used in a manner that is more expansive than its ordinary and customary meaning, the term should be given its ordinary and customary meaning plus the additional expansive meaning, or (b) if a term has been explicitly defined to have a different meaning by reciting the term followed by the phrase "as used in this document shall mean" or similar language (e.g., "this term means," "this term is defined as," "for the purposes of this disclosure this term shall mean," etc.).

[0097] In this disclosure, the term "closed system" can refer to a system that is sealed from the external environment of the system such that the internal and external environments of the system do not interact or have minimized interactions.

[0098] In this disclosure, the term "aseptic" can refer to the absence of microorganisms.

[0099] In this disclosure, the term "sterile" can refer to the absence of all microorganisms.

[0100] References to specific examples, use of "i.e.," use of the word "invention," etc., are not meant to invoke exception (b) or otherwise restrict the scope of the recited claim terms. Other than situations where exception (b) applies, nothing contained in this document should be considered a disclaimer or disavowal of claim scope.

[0101] The subject matter recited in the claims is not coextensive with and should not be interpreted to be coextensive with any embodiment, feature, or combination of features shown in this document. This is true even if only a single embodiment of the feature or combination of features is illustrated and described in this document. Thus, the appended claims should be given their broadest interpretation in view of the prior art and the meaning of the claim terms.

[0102] Spatial or directional terms, such as "left," "right," "front," "back," "top," "bottom," and the like, relate to the subject matter as it is shown in the drawings. However, it is to be understood that the described subject matter can assume various alternative orientations and, accordingly, such terms are not to be considered as limiting.

[0103] Articles such as "the," "a," and "an" can connote the singular or plural. Also, the word "or" when used without a preceding "either" (or other similar language indicating that "or" is unequivocally meant to be exclusive—e.g., only one of x or y, etc.) shall be interpreted to be inclusive (e.g., "x or y" means one or both x or y).

[0104] The term "and/or" shall also be interpreted to be inclusive (e.g., "x and/or y" means one or both x or y). In situations where "and/or" or "or" are used as a conjunction for a group of three or more items, the group should be interpreted to include one item alone, all the items together, or any combination or number of the items. Moreover, terms used in the specification and claims such as have, having, include, and including should be construed to be synonymous with the terms comprise and comprising.

[0105] Unless otherwise indicated, all numbers or expressions, such as those expressing dimensions, physical characteristics, and the like, used in the specification (other than the claims) are understood to be modified in all instances by the term "approximately." At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the claims, each numerical parameter recited in the specification or claims which is modified by the term "approximately" should at least be construed in light of the number of recited significant digits and by applying ordinary rounding techniques.

[0106] All disclosed ranges are to be understood to encompass and provide support for claims that recite any and all subranges or any and all individual values subsumed by each range. For example, a stated range of 1 to 10 should be considered to include and provide support for claims that recite any and all subranges or individual values that are between and/or inclusive of the minimum value of 1 and the maximum value of 10; that is, all subranges beginning with a minimum value of 1 or more and ending with a maximum value of 10 or less (e.g., 5.5 to 10, 2.34 to 3.56, and so forth) or any values from 1 to 10 (e.g., 3, 5.8, 9.9994, and so forth).

[0107] All disclosed numerical values are to be understood as being variable from 0-100% in either direction and thus provide support for claims that recite such values or any and all ranges or subranges that can be formed by such values. For example, a stated numerical value of 8 should be understood to vary from 0 to 16 (100% in either direction) and provide support for claims that recite the range itself (e.g., 0 to 16), any subrange within the range (e.g., 2 to 12.5) or any individual value within that range (e.g., 15.2).

[0108] The operations presented in this document are not inherently related to any particular apparatus. Various general-purpose systems can also be used in accordance with the teachings in this document, or it can prove convenient to construct more specialized apparatuses to perform the required method steps. The required structure for a variety of these systems will be apparent to those of skill in the art, along with equivalent variations.

[0109] The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same can also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

What is claimed is:

1. A system for product delivery, comprising:
- a container; and
- a product delivery device comprising:
  - a first piece comprising a collar piece and a distal lip;
  - a second piece comprising:
    - a conical section with a wide end and a narrow end; and
    - a stopper comprising a seating lip;
  - an inlet port; and
  - an outlet port flush with the narrow end of the conical section;
- wherein the container is attached to the first piece of the product delivery device; and
- wherein the system is configured to transfer product from the container through the conical section of the second piece, and then through the outlet port of the second piece.

2. The system of claim **1**, wherein the first piece further comprises a securement feature on an inner surface of the collar piece, configured to attach to the container.

3. The system of claim **2**, wherein the container has threads, and the securement feature is engaged with the threads.

4. The system of claim **3**, wherein the seating lip is compressed between the distal lip and an end of the container that is closest to the threads.

5. The system of claim **4**, wherein the system is a closed system to the environment.

6. The system of claim **5**, wherein the stopper is a flexible material, and the collar piece is a rigid material.

7. The system of claim **1**, wherein the inlet port and the outlet port have external barbs.

8. The system of claim **1**, wherein the container further comprises a product, the product comprising a dry free flowing powder.

9. The system of claim **8**, wherein the product comprises dried, dissolvable microcarriers.

10. A system for aseptic dry product delivery, comprising:
- a container at least partially filled with an aseptic dry product;
- a product delivery device comprising:
  - a first piece comprising a collar piece and a distal lip;
  - a second piece comprising:
    - a conical section with a wide end and a narrow end; and
    - a stopper comprising a seating lip;
  - an inlet port; and
  - an outlet port flush with the narrow end of the conical section;
- a pressure source connected to the inlet port; and
- a receiving vessel configured to receive the aseptic dry product;

wherein the container is attached to the first piece of the product delivery device; and

wherein the system is configured to transfer product from the container through the conical section of the second piece, and then through the outlet port of the second piece to the receiving vessel.

11. The system of claim **10**, wherein the system is a closed system to the environment.

12. The system of claim **10**, wherein the inlet port is connected to a dip tube that extends from the inlet port to a side of the container opposite of the inlet port.

13. The system of claim **10**, wherein an end of the inlet port that faces the container extends into an interior portion of the container.

14. The system of claim **10**, further comprising:

a first tube connected to the inlet port;

a first clamp configured to pinch the first tube;

a second tube connected to the outlet port; and

a second clamp configured to pinch the second tube.

15. The system of claim **14**, wherein the first tube comprises a vent and a vent filter.

16. The system of claim **14**, wherein the container comprises a vent.

17. The system of claim **10**, wherein the inlet port and the outlet port have external barbs.

18. The system of claim **10**, wherein the receiving vessel is configured to hydrate the dry dissolvable microcarriers.

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